

Bioleach PFD Index

Everything the drawing deliberately leaves off: sizing basis per tag, full stream compositions, and numbered notes. Circled numbers on the PFD are footnotes — they resolve here, numbered continuously across blocks. Drawing: Black Mass Bioleach PFD.

Design basis

THROUGHPUT 10,000 t/yr black mass · 8,000 h/yr · 1,250 kg/h	FEED GRADE (WT %) Ni 18 · Co 8 · Mn 5 · Li 3.5 · Cu 2 · Al 3 · other 60.5	LIXIVIANT Gluconic acid (<i>G. oxydans</i> broth) + FeSO ₄ reductant. CHARGE-BALANCED: GLUCONATE SUPPLIES THE WHOLE ANION BUDGET. ~900 mM in the leach liquor (≈ 5.0 t/h gluconic acid: the metals dissolved plus a 15 % free-acid margin so the leach has a finite pH, ~4.8; fixed by feed chemistry, not by pulp), delivered as a 150 g/L fed-batch broth — the literature high for <i>G. oxydans</i> ^[12] ; INL lab broth was 80 g/L). No mineral acid in the leach; see the acid-budget note below.
LEACH CONDITIONS 55 °C · 30 h · INL 2022 (INL/JOU- 22-70203)	PULP DENSITY 4.2 % — the highest pulp at which a 150 g/L broth can carry the required gluconate without negative make-up water. The reference work demonstrated 2.5 %.	LEACH RECOVERY Co 86 · Ni 84 · Li 100 · Mn 100 %. Downstream losses applied unit by unit.

Acid budget. Dissolved metals need ≈ 25 keq/h of counter-anions; a pulp-scaled reference basis (300 mM gluconate at 10 %) supplied ≈ 6 keq/h, a 4× deficit that PHREEQC ^[19] confirmed independently (cation excess 1.7 eq/L). The charge-balanced basis closes it with gluconate alone: 896 mM at 4.2 % pulp (15 % free-acid margin), charge ratio 1.13. Downstream base demand is computed as what it takes to hold each unit's pH set-point in the ~1 M gluconate buffer (Fe/Al 5.5, Mn SX 4.2, Co SX 5.5, Ni SX 6.5), not one NaOH per gluconate — the leach already deprotonated the gluconate that balances the metals. Consequences carried through the flowsheet: R-202 575 m³ per vessel, glucose 4.8 t/h, NaOH 0.84 t/h, and a barren liquor that is ~240 g/L sodium gluconate — which is why lithium is taken as Li₃PO₄ before, not after, the evaporator (notes 28–29). Options A/B/C and the reasoning are in `Process/Charge Balance Finding.md`.

Mass balance and sizing: `G-mass-balance/mass_balance.py`. Speciation and precipitation thermodynamics: `F-thermodynamics/thermo_speciation.py` (PHREEQC via phreeqpython, USGS minteq.v4 + gluconate constants added; output in `F-thermodynamics/thermo_results.txt`). The **Sizing** and **Streams** tables in every block are its output (`python3 mass_balance.py --html N`); the equipment tables are hand-written for material and design notes. Stream notes that were in the tables now live in the numbered notes.

Block 1

Area 100 · Black mass receiving

EQUIPMENT

TAG	SERVICE	TYPE	SIZING BASIS	MATERIAL / NOTES
V-101	Black mass receiving silo	Cone-bottom vertical silo, bulk-bag unloader	60 t (48 h hold-up)	CS · N ₂ blanket — fine graphite + residual LiPF ₆ (dust ignition, HF off-gas)
F-101	Silo dust collector	Pulse-jet baghouse	vent + displacement air	Fines return to V-101; clean air to atmosphere

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
V-101	Black mass receiving silo	60.0 t	48 h hold-up

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

NO.	FROM → TO	PHASE	TOTAL KG/H	NI	CO	MN	LI	CU	AL	OTHER	T °C
S-101	OSBL → V-101	dry solid	1,250	225	100	62.5	43.8	25.0	37.5	756	25
S-102	V-101 vent → F-101	gas+dust	0	0	0	0	0	0	0	0	25

NOTES — CIRCLED NUMBERS ON THE DRAWING

- 1

Battery pretreatment is off-plot (OSBL). Upstream of S-101: discharge → shred under inert gas → binder / electrolyte removal (thermal or solvent) → sieve with Cu/Al foil rejection. Only black mass (<200 μm) crosses the battery limit. Out of scope, but the wash and foil-reject efficiency set the F, Cu and Al load on the effluent trains (Areas 400/500) — the one upstream variable this flowsheet is sensitive to.
- 2

N₂ blanket on V-101. Black mass is fine graphite carrying residual LiPF₆ — an ignition and HF-off-gas concern. Shown as a utility stub rather than omitted.
- 3

48 h hold-up (60 t). Decouples truck deliveries from a continuous leach. Sized at the 1,250 kg/h feed rate.

Block 2 Area 100 · Metering and slurry make-up

EQUIPMENT

TAG	SERVICE	TYPE	SIZING BASIS	MATERIAL / NOTES
W-101	Black mass metering	Loss-in-weight screw feeder	1,250 kg/h · WIC	Sets the plant feed rate; only place the 10,000 t/yr basis is enforced
T-101	Slurry make-up tank	Vertical, dished bottom, pitched-blade turbine (axial, solids suspension)	22 m ³ · 2 h at 10.6 t/h	Rubber-lined CS · surge for a bag change / feeder trip · level to be trended (LIC not drawn)
P-101 A/B	Feed slurry transfer	Progressive-cavity, installed spare	10.6 t/h · ~9.8 m ³ /h	Abrasive graphite slurry; PC pump avoids the erosion a centrifugal impeller sees at 10 % solids

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
T-101	Slurry make-up tank	2.6 m ³	2 h at S-105
P-101 A/B	Feed slurry pump	1.3 m ³ /h	S-105 volumetric

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

NO.	FROM → TO	PHASE	TOTAL KG/H	NI	CO	MN	LI	CU	AL	OTHER	T °C
S-103	W-101 → T-101	dry solid	1,250	225	100	62.5	43.8	25.0	37.5	756	25
S-104	Sh3/4 recycle + fresh → T-101	liquid	141	0	0	0	0	0	0	0	25
S-105	P-101 → R-301	slurry	1,391	225	100	62.5	43.8	25.0	37.5	756	25

NOTES — CIRCLED NUMBERS ON THE DRAWING

- 4 **W-101 loss-in-weight feeder.** The one instrument that enforces the 1,250 kg/h basis. Everything downstream is sized from what this feeder delivers, so it carries the WIC (weight indicating controller) rather than a simple rotary valve.
- 5 **S-104 make-up water is a blend.** Recycled residue-wash filtrate (Area 300) and treated raffinate (Area 400), topped up with fresh water. Recycle fraction is a design variable; a bleed on the raffinate loop (Area 500) controls sulfate and Na/Ca build-up — without it the loop is not closed, only slowly dying.
- 6 **Pulp-density control.** A density transmitter on the P-101 discharge (DIC) trims the S-104 valve. This is the loop that guards the 7 % basis: solids are fixed by W-101, so water is the only free variable.
- 7 **Progressive-cavity pumps.** Graphite-laden slurry is abrasive; a PC pump moves it at constant flow with low shear and tolerates the solids a centrifugal would erode on. A/B = installed spare.

DESIGN CHOICE

- Slurry in water, dose biolixiviant at the reactor.** Keeps T-101 out of the Fe(II)/gluconate chemistry and lets pulp density and lixiviant dose be controlled independently. The alternative — slurrying straight into gluconate broth — saves nothing but the water line and couples two controls that should stay separate.

Block 3 Area 200 · Biolixiviant fermentation (Sheet 2)

EQUIPMENT

Tag	Service	Type	Sizing Basis	Material / Notes
V-201	Glucose syrup storage	Vertical tank	640 kg/h glucose · 7 d as 70 % syrup	SS304. Corn-stover hydrolysate is the INL carbon source; glucose syrup is the drop-in for a first plant
T-201	Medium make-up	Agitated, jacketed (steam-sterilizable)	1 batch ≈ 845 m³ · 24 h	Pikovskaya salts + glucose + returned raffinate + fresh water. N/P here is the origin of the nutrient duty on Sheet 5
R-201A	Pre-seed fermenter	Stirred, sparged, jacketed	50 L working	Inoculated from lab shake flask (off-sheet)
R-201B	Seed fermenter	Stirred, sparged, jacketed	5 m³ working	10 % inoculum from R-201A
R-202 A/B	Production fermenter	Stirred (2× Rushton), ring sparger, jacketed	~960 m³ each · ~800 m³ working · 24 h · staggered	SS316L · 30 °C, pH 5.5→3 uncontrolled · SIP/CIP
K-201 / F-202	Sterile air	Rotary-lobe blower + 0.2 µm depth filter	0.5 vvm on the R-202 working volume (sizing table shows the scaled figure)	One header serves all four vessels
F-201	Cell removal	0.22 µm crossflow (or disc-stack centrifuge)	8.3 m³/h broth	INL used 0.22 µm filtration; centrifuge is the industrial substitute
T-202	Biolixiviant hold	Vertical, closed	12 h at 8.3 m³/h	Turns two batch harvests into a continuous 2 m³/h dose
P-201	Biolixiviant dose	Centrifugal (clean liquid)	8.3 m³/h · FIC to R-301	Dose rate is the gluconate control on the leach

SIZING – GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
V-201	Glucose syrup storage	861 m3	7 d of S-201 as 70 % syrup
T-201	Medium make-up	845 m3/batch	1 batch of S-202
R-201A	Pre-seed fermenter	80.2 L	1/100 of R-201B (1 % inoculum)
R-201B	Seed fermenter	80.2 m3	1/10 of R-202 working
R-202 A/B	Production fermenter	962 m3 each	24 h of broth, x1.2 headspace
K-201	Sterile air blower	24,053 m3/h	0.5 vvm on R-202 working volume
F-201	Cell removal	33.4 m3/h	S-209 volumetric
T-202	Biolixiviant hold	401 m3	12 h at S-211
P-201	Biolixiviant dose pump	33.4 m3/h	S-212 volumetric

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

[illegible]

NO.	FROM → TO	PHASE	TOTAL KG/H	H ₂ O	GLUCONIC ACID	GLUCOSE	ORGANICS / CELLS	AIR	T °C							
S-204	tee → Sh 1 make-up (S- 104)	liquid	4,010	4,010	0	0	0	0	0	0	0	0	0	0	0	25
S-205	K-201/F-202 → R-201/202	gas	28,864	0	0	0	0	0	0	0	0	0	0	28,864	0	25
S-206	R-201/202 → atm	gas	28,455	0	0	0	0	0	0	0	0	0	0	28,455	0	30
S-207	R-201A → R- 201B	broth	490	450	40.0	0	0	0	0	0	0	0	0	0	0	30
S-208	R-201B → R- 202 A/B	broth	4,900	4,500	400	0	0	0	0	0	0	0	0	0	0	30
S-209	R-202 A/B → F- 201	broth	33,437	28,401	5,011	5.0	20.0	0	0	0	0	0	0	0	0	30
S-210	F-201 → Sh 5 effluent	wet cake	40.0	30.0	0	0	10.0	0	0	0	0	0	0	0	0	30
S-211	F-201 → T-202	liquid	33,397	28,371	5,011	5.0	10.0	0	0	0	0	0	0	0	0	30
S-212	P-201 → R-301	liquid	33,397	28,371	5,011	5.0	10.0	0	0	0	0	0	0	0	0	30

NOTES – CIRCLED NUMBERS ON THE DRAWING

- 8 **Seed train.** Lab shake flask (off-sheet) → R-201A 50 L → R-201B 5 m³ → R-202 50 m³, 10 % inoculum at each step. Same shape as a pharma cell-culture train, at a fraction of the cost: the product is a dilute acid, not an injectable, so the plant is *aseptic* rather than sterile-critical, and gluconic acid at pH ~3 self-selects against most contaminants. Brewery-grade stainless with SIP/CIP, not GMP.
- 9 **Staggered batch, not continuous.** (1) Product inhibition — *G. oxydans* stalls as gluconate accumulates and pH falls; a continuous fermenter runs at the outlet condition and can only hold low titer, which would multiply the broth volume dosed at R-301 and break the pulp-density basis. (2) Containment — a contaminated batch costs 24 h of one vessel; a contaminated continuous fermenter starves the leach. (3) Precedent — industrial gluconic acid has been batch/fed-batch since the 1950s and INL's demonstration was batch; no continuous reference plant exists. (4) The leach still sees continuous feed: A/B offset 12 h plus T-202's 12 h hold. (5) The charge-balanced gluconate requirement at a 150 g/L broth makes the production fermenter ~960 m³ per vessel — large industrial fermenter scale (citric/gluconic plants run 100–300 m³ for real citric/gluconic plants), where the batch-vs-continuous case should be revisited.
- 10 **One sterile-air train.** K-201 + F-202 0.2 µm feed all four vessels from one header. O₂ demand is ½ mol per glucose — ~410 kg/h, and the 0.5 vvm rate on the ~800 m³ working volume (~24,000 m³/h air) supplies several times the stoichiometric O₂; mass transfer, not stoichiometry, still sets the rate.
- 11 **Condensate return, not raffinate (revised at Sheet 5).** The SX raffinate leaves Area 500 as a sodium-gluconate brine (~640 kg/h Na, ~5.0 t/h gluconate) that the fermenter cannot take, so the water that comes back is the *evaporator and crystalliser condensate* (S-517, ~22 t/h, clean). It supplies the medium water here and the surplus (S-204) goes to T-101 make-up on Sheet 1. Nutrients and gluconate therefore do *not* recycle — they leave in the brine bleed (S-516) to Sheet 6. Sheet 6 now closes the balance: condensate ~10.4 t/h covers the S-104 make-up and most of the medium, leaving ~6.9 t/h net fresh water. Condensing the D-601 exhaust (~7.8 t/h) would close it entirely (note 32).
- 12 **Broth water counts toward pulp density.** S-212 brings ~28 t/h of water into R-301 (33 m³/h of 150 g/L broth), so S-104 make-up water on Sheet 1 is only ~1,700 kg/h against a 16.6 t/h liquor; T-101 is now a small make-up tank rather than the ~43 % solids paste the 7 % basis required and the 4.2 % basis is reached in the reactor. Most of the leach water is broth — the fermenter is the water plant.

Block 4 Area 300 · Batch leach and solid/liquid separation (Sheet 3 of 6)

EQUIPMENT

TAG	SERVICE	TYPE	MATERIAL / NOTES
E-301	Feed slurry preheater	Spiral or wide-gap plate, steam	Ambient → 55 °C. Jackets on R-301 hold temperature; E-301 does the bulk duty so the batch does not spend its first hours warming up
V-301 / W-301	FeSO ₄ ·7H ₂ O silo + loss-in-weight feeder	Cone silo, screw feeder	Dry dosing into the fill line — no water added, pulp basis intact. Ratio-controlled to W-101 (kg Fe per kg feed)
R-301 A-D	Batch leach tanks	Agitated (2× pitched-blade), jacketed, closed	Rubber-lined CS; 55 °C; pH ~2.5–3 uncontrolled (set by gluconate). Cycle: 12.5 h fill · 30 h leach · 7.5 h drain = 50 h, four tanks offset 12.5 h
SC-301	Vent scrubber	Packed column, NaOH recirculation	CO ₂ from residual Li ₂ CO ₃ , HF traces from electrolyte carry-over. Blowdown to Sheet 6 (S-304 → T-601)
T-303 / P-303	Leached-slurry surge + filter feed	Agitated tank, PC pump	Turns a 7.5 h batch drain into steady 12 m ³ /h to the belt filter — the price of batch tanks
F-301	Residue filter	Horizontal vacuum belt, 2-stage countercurrent wash	Strong filtrate → T-301; wash filtrate → T-302. Graphite fines may blind the cloth: filterability test is the first lab question this sheet raises
T-301 / P-301	PLS tank + pump	Closed tank, centrifugal	Pregnant leach solution to Fe removal (Sheet 4)
T-302 / P-302	Wash-filtrate tank + pump	Closed tank, centrifugal	Returns to T-101 as part of S-104 — carries ~2 % of the leached Co back into the leach, which is a recycle, not a loss

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
V-301	FeSO ₄ .7H ₂ O silo	59.2 t	7 d of S-301
E-301	Slurry preheater	1,015 kW	S-302 x 3.5 kJ/kgK x (55-25)
R-301 A-D	Batch leach tank	483 m ³ each	12.5 h fill at S-302, x1.2 headspace
T-303	Leached-slurry surge	258 m ³	8 h at S-305 (drain 150 m ³ in 7.5 h vs filter 12 m ³ /h)
F-301	Belt filter	7.8 m ²	dry residue / 100 kg/m ² .h
T-301	PLS tank	136 m ³	4 h at S-309
T-302	Wash-filtrate tank	6.3 m ³	4 h at S-311
SC-301	Vent scrubber	116 m ³ /h gas	S-303 + displacement air

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	T °C
S-301	V-301/W-301 → R-301	dry solid	353	0	0	0	0	0	0	0	160	70.8	122	0	25
S-302	E-301 tee → R-301	slurry	34,788	225	100	62.5	43.8	25.0	37.5	756	28,512	0	0	5,011	55
S-303	R-301 → SC-301	gas	41.6	0	0	0	0	0	0	0	0	0	0	0	55
S-304	SC-301 → Sh 5 effluent	liquid	91.6	0	0	0	0	0	0	0	50.0	0	0	0	25
S-305	R-301 → T-303	slurry	35,099	225	100	62.5	43.8	25.0	37.5	715	28,672	70.8	122	5,011	55
S-306	P-303 → F-301	slurry	35,099	225	100	62.5	43.8	25.0	37.5	715	28,672	70.8	122	5,011	55
S-307	process water → F-301	liquid	1,569	0	0	0	0	0	0	0	1,569	0	0	0	25
S-308	F-301 → Sh 6 residue bunker X-601	wet cake	1,121	36.1	14.0	0	0	5.0	15.0	715	334	0	0.1	2.5	40

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	T °C
S-309	F-301 strong → T-301	liquid	33,978	187	85.2	61.9	43.3	19.8	22.3	0	28,391	70.1	121	4,962	45
S-310	P-301 → Sh 4 (Fe removal)	liquid	33,978	187	85.2	61.9	43.3	19.8	22.3	0	28,391	70.1	121	4,962	45
S-311	F-301 wash → T-302	liquid	1,569	1.8	0.8	0.6	0.4	0.2	0.2	0	1,517	0.7	1.1	46.7	40
S-312	P-302 → Sh 1 T-101 (S-104)	liquid	1,569	1.8	0.8	0.6	0.4	0.2	0.2	0	1,517	0.7	1.1	46.7	40

NOTES – CIRCLED NUMBERS ON THE DRAWING

- 13 **Batch leach, staggered.** INL's 30 h / 55 °C / 86 % Co figure was measured in a batch vessel; a batch tank reproduces that condition exactly, where a CSTR train only approximates it through a residence-time distribution. Four tanks offset by 12.5 h give the continuous filter a steady feed via T-303. Same argument as the fermenters (note ⑩): batch where the kinetics were measured, continuous where the equipment wants it.
- 14 **Iron enters dry.** ~352 kg/h FeSO₄·7H₂O — 4.1 kg per kg Co recovered (INL 3.75–4.51). At 28 % of the black-mass feed rate it is the single largest reagent, 10 % of OpEx, and every kilogram leaves as Fe(III) sludge on Sheet 4. Dosed as dry salt into the fill line so no water is added; a 20 % solution would have cut S-104 by another 1.4 t/h.
- 15 **Off-gas.** Carbonate in the black mass (Li₂CO₃, SEI) releases CO₂ on acid contact; residual LiPF₆ can release HF. A small caustic scrubber is drawn rather than an open vent because of the HF, not the CO₂. CO₂ is a placeholder (30 % of Li as carbonate).
- 16 **Residue is the largest single loss.** ~830 kg/h dry: graphite and binder plus the unleached 14 % of the Co (14.1 kg/h) and 16 % of the Ni. To Sheet 5 as waste for this flowsheet; graphite recovery and a second-stage residue leach are the two obvious upgrades, both needing data INL does not give.
- 17 **Wash filtrate closes the first water loop.** S-312 (~1.6 t/h) is the "Sh 3" arrow on Sheet 1 — about 90 % of the S-104 make-up (make-up is small because the broth is most of the water). It carries the PLS displaced from the cake, so a few kg/h of dissolved Co/Ni cycle back through the leach at steady state. Not a loss, but the balance here is single-pass; steady-state build-up is a later refinement.
- 18 **Cu and Al leach fractions — literature-anchored, not measured.** INL does not report them. Gluconic acid on its own barely dissolves copper (0.3 % Cu, 9.7 % Fe, 51.9 % Al — Lerchbammer et al. 2025 ^[11], an H₂O₂ system), but once Fe³⁺ is present metallic Cu in black mass is consumed as a reductant (Porvali et al. 2020 ^[22]; Partinen et al. 2024), and organic-acid bioleaching reaches 100 % Cu with 65–75 % Al (Horeh et al. 2016 ^[8]; Bahaloo-Horeh et al. 2017 ^[9]). Basis: **Cu 80 %, Al 60 %**; sensitivity Cu 0.3–100 %. Upside not taken: Cu's electrons would offset ~half the FeSO₄ dose, but INL's 4.1 kg/kg ratio was measured on Cu-bearing black mass, so the credit is assumed already inside it. Both fractions drive Sheet 4 (Cu poisons Co/Ni recovery; Al makes gelatinous hydroxide). Thermodynamic ceiling from `thermo_speciation.py`: see note ⑩.
- 19 **Thermodynamics of the PLS (PHREEQC, minteq.v4 + gluconate; leach liquor from the balance).** Liquor from the balance (Co 4.0, Ni 8.7, Mn 2.9, Li 2.0, Cu 0.9, Al 1.0, Fe 3.3 g/L; gluconate 200 g/L ≈ 1.0 M; 55 °C, Fe as Fe(III)). **(0) Charge balance is pH-dependent.** Gluconic acid has pKa 3.86, so at the INL leach pH of 2.5 only a few % of the 1 M gluconate is the anion and PHREEQC still reports a 1.0 eq/L cation excess. The balanced liquor therefore cannot sit at pH 2.5: as metals dissolve and consume protons the pH rises to ~4.5–5, where gluconate is deprotonated and the budget closes. **Leach pH is now a design variable, not the INL number. (a) Iron and aluminium drop out in the leach itself.** With 1 M gluconate goethite is complete from pH 3 (Fe dissolved 0.2 %) and gibbsite takes Al from pH 4 (55 % out at 4.0, 91 % at 4.5, 99 % at 5.5); Co, Ni, Cu, Mn stay 100 % dissolved through pH 8, held by gluconate chelation. Most of the Fe/Al removal happens in R-301 and reports to the leach residue (S-308), and T-402's duty shrinks to a polish — but the Fe(III) oxidant is largely solid above pH 3 (only 3–45 % Fe(III)-gluconate in solution), which questions the FeSO₄ reductant chemistry at that pH. **(b) Cu now stays dissolved to pH 8** (was 5 % on the 300 mM basis): cementation (T-401) remains the right removal step. **(c) Fe(III)-gluconate log K** still controls whether Fe is held in solution: ≤ 8 → Fe precipitates, 10 → 100 % dissolved at pH 3.5. Still the single most valuable lab measurement. *Caveats:* I ≈ 1 M is beyond the Davies model; 25 °C gluconate constants; goethite as control phase; equilibrium only. Output: `thermo_results.txt`.

Block 5 Area 400 · Impurity removal: Cu → Fe/Al → Mn (Sheet 4)

EQUIPMENT

TAG	SERVICE	TYPE	MATERIAL / NOTES
V-401 / W-401	Iron powder storage + feeder	Hopper, screw feeder	1.3× stoichiometric on Cu; ratio-controlled to the PLS Cu assay
T-401	Copper cementation	Agitated, closed (H ₂ vent)	pH 2.5 as received; Cu ²⁺ + Fe ⁰ → Cu ⁰ + Fe ²⁺ . Chosen over sulfide precipitation or SX — see note 20
F-401	Cement copper filter	Small filter press / candle filter	~30 kg/h wet cake, ~64 % Cu — a salable intermediate
T-402	Fe oxidation + Fe/Al precipitation	Agitated, air-sparged, NaOH dosed, pHc	Fe ²⁺ → Fe ³⁺ with air, then NaOH to pH 5.5 → goethite + gibbsite (thermo note 21). 50 °C. NaOH not lime — note 21
K-401	Oxidation air	Rotary-lobe blower	2× stoichiometric O ₂ on Fe ²⁺
F-402	Fe/Al cake filter	Membrane filter press	Gelatinous Al(OH) ₃ blinds cloths — the press, not a belt. Cake to Sheet 6; carries occluded Co/Ni (note 22)
X-401 / 402 / 403	Mn solvent extraction	Mixer-settlers: 2 extraction, 1 scrub, 2 strip	20 % D2EHPA in kerosene, O/A 1, pH 3.0 held with NaOH; scrub with water; strip with H ₂ SO ₄ . Also scavenges residual Fe/Al (note 23)
T-404	MnSO ₄ strip liquor	Closed tank	~100 g/L Mn. Product as liquor or to a crystalliser (future)
T-405 / P-405	Purified liquor to Sheet 5	Closed tank, centrifugal	Co 80 · Ni 180 · Li 43 kg/h; Mn 1.2, Cu 0.2, Fe/Al nil

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
T-401	Cementation reactor	34.0 m3	1 h at S-401
F-401	Cement Cu filter	24.8 kg/h dry	S-403 dry solids
T-402	Fe/Al oxidation + neutralisation	81.5 m3	2 h at S-404, x1.2
K-401	Oxidation air blower	89.5 m3/h	S-405 at 1.2 kg/m3
F-402	Fe/Al cake filter press	13.9 m2	dry cake at 15 kg/m2.h
X-401/402/403	D2EHPA mixer-settlers (2E + 1S + 2St)	11.4 m3 mixer each	10 min mixer at O/A 1, S-409 + organic
T-404	MnSO4 strip liquor tank	14.6 m3	24 h at S-413
T-405	Purified liquor tank	137 m3	4 h at S-414

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	NA	T °C
S-401	Sh 3 PLS → T-401	liquid	33,978	187	85.2	61.9	43.3	19.8	22.3	0	28,391	70.1	121	4,962	0	45
S-402	V-401/W-401 → T-401	dry solid	22.4	0	0	0	0	0	0	0	0	22.4	0	0	0	25
S-403	F-401 → Cu cement product	wet cake	31.0	0	0	0	0	19.6	0	0	6.2	5.2	0	0	0	25
S-404	F-401 → T-402	liquid	33,969	187	85.2	61.9	43.3	0.2	22.3	0	28,385	87.4	121	4,962	0	45
S-405	K-401 → T-402	gas	107	0	0	0	0	0	0	0	0	0	0	0	0	25
S-406	NaOH 50 % → T-402	liquid	627	0	0	0	0	0	0	0	398	0	0	0	229	25

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	NA	T °C
S-407	T-402 → F-402	slurry	34,718	183	83.5	61.9	43.3	0.2	0	209	28,811	0	121	4,962	229	50
S-408	F-402 → Sh 6 Fe/Al cake	wet cake	522	3.7	1.7	0	0	0	0	203	313	0	0	0	0	25
S-409	F-402 → X-401 Mn SX	liquid	34,196	183	83.5	61.9	43.3	0.2	0	0	28,497	0	121	4,962	229	45
S-410	NaOH 50 % → X-401	liquid	0	0	0	0	0	0	0	0	0	0	0	0	0	25
S-411	X-401 → X-403 (organic loop) → internal	organic	27,357	0	0	0	0	0	0	0	0	0	0	0	0	25
S-412	H2SO4 98 % → X-403	liquid	108	0	0	0	0	0	0	0	2.2	0	106	0	0	25
S-413	X-403 strip → T-404 MnSO4 product	liquid	608	0	1.7	60.6	0	0	0	0	440	0	106	0	0	25
S-414	X-401 raffinate → T-405 → Sh 5	liquid	34,133	183	81.8	1.2	43.3	0.2	0	0	28,497	0	121	4,962	229	40

NOTES – CIRCLED NUMBERS ON THE DRAWING

- 20 **Cementation puts iron back in.** Every mole of Cu removed adds a mole of Fe²⁺ (~17 kg/h) to a liquor that is about to be de-ironed, and the excess powder (30 %) reports to the cement cake. Cost: an air-oxidation duty in T-402 and ~20 % more goethite. Benefit: no H₂S handling, no SX circuit for 20 kg/h of copper, and the cement is salable. The alternative sulfide route (NaHS at pH 2–3) would avoid the iron but needs a scrubbed vent.
- 21 **NaOH, not lime, for the Fe/Al step.** Dose is set by the pH target in the gluconate buffer: after the leach the liquor already sits near pH 4.8, so reaching 5.5 while dropping Fe³⁺ and Al³⁺ as hydroxides takes ~390 kg/h NaOH (not the ~1.2 t/h a one-NaOH-per-gluconate count gives). Mn SX at pH 4.2 needs none; Co and Ni SX take ~190 and ~260 kg/h to hold 5.5 and 6.5 as the extractants release protons. This is a gluconate liquor with only ~6 g/L sulfate (from FeSO₄), so lime would strip all of it as gypsum and leave ~90 kg/h of dissolved Ca — which D2EHPA extracts ahead of Mn and which would contaminate every downstream product. Sodium goes to the raffinate bleed instead. Lime is kept for the effluent sheet, where gypsum is wanted.
- 22 **Co/Ni occlusion in the Fe/Al cake — placeholder 2 % each.** Goethite/gibbsite co-precipitation loss is real but unmeasured here; 1.7 kg/h Co is ~\$1.5 M/yr. The number to test first in the lab, together with cake filterability.
- 23 **D2EHPA runs at pH 3.0 and doubles as a scavenger.** Extraction order Fe³⁺ > Al³⁺ > Ca > Mn > Co > Ni: whatever Fe/Al passes F-402 is taken here and leaves with the strip liquor. Co co-extraction is held to ~2 % by keeping pH ≤ 3; above ~3.5 cobalt follows manganese. Two extraction stages for 98 % Mn; scrub stage returns entrained Co/Ni.
- 24 **MnSO₄ is a by-product, not a headline.** ~60 kg/h Mn as ~100 g/L strip liquor. Sold as liquor to a Mn refiner or crystallised later; not in the revenue basis. Carries ~1.6 kg/h Co that the scrub stage should reduce.

Block 6 Area 500 · Products and water return (Sheet 5)

EQUIPMENT

TAG	SERVICE	TYPE	MATERIAL / NOTES
X-501 / 502 / 503	Cobalt solvent extraction	Mixer-settlers: 3 extraction, 2 scrub, 2 strip	Cyanex 272 ^[15] , saponified with NaOH; pH 5.5–6.0; O/A 1. Ni co-extraction ~1 %, scrubbed back (note 25)
X-504 / C-501 / D-501	CoSO ₄ ·7H ₂ O	Evaporative crystalliser, pusher centrifuge, fluid-bed dryer	Strip liquor ~100 g/L Co; 86 % per-pass [14] with the mother liquor recycled; only a 5 % bleed leaves as S-518, so overall recovery is 99.2 %. Product is the hydrated sulfate (note 27)
X-511 / 512 / 513	Nickel solvent extraction	Mixer-settlers: 3 extraction, 1 scrub, 2 strip	Versatic 10 (neodecanoic acid), pH 6.5–7; extracts Ni ahead of Li/Na (note 26)
X-514 / C-502 / D-502	NiSO ₄ ·6H ₂ O	Evaporative crystalliser, centrifuge, dryer	As above; purge S-519
T-501 / F-501	Lithium phosphate	Agitated, jacketed 60 °C; pressure filter with water wash	Na ₃ PO ₄ 10 % excess; Li ₃ PO ₄ solubility ~0.39 g/L so it precipitates straight from the ~1.9 g/L barren — no evaporation ahead of it (note 29)
E-501	Water recovery	Mechanical vapour recompression evaporator	Li-free barren 22.7 t/h → 13.7 t/h brine at ~500 g/L Na-gluconate; 9 t/h condensate back to Sheets 1–3. Largest energy user after the fermenter air (note 28)
Condensate header	Water return	—	S-517 to Sheets 1–3: medium water, T-101 make-up, F-301 wash. Closes the water loop Sheet 2 note 11 assumed

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
X-501/502/503	Co SX mixer-settlers (3E + 2S + 2St)	11.4 m3 mixer each	10 min mixer at O/A 1
X-504	CoSO4 crystalliser (evaporative)	0.4 t/h H2O	water evaporated
X-511/512/513	Ni SX mixer-settlers (3E + 1S + 2St)	11.4 m3 mixer each	10 min mixer at O/A 1
X-514	NiSO4 crystalliser (evaporative)	0.9 t/h H2O	water evaporated
E-501	Barren evaporator (MVR), water recovery	20.9 t/h H2O	to 500 g/L Na-gluconate
T-501	Li3PO4 precipitation	34.6 m3	1 h at barren flow, 60 C
D-501/D-502	Product dryers	1,174 kg/h	CoSO4 + NiSO4 crystals

STREAMS — GENERATED BY MASS_BALANCE.PY (KG/H)

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	NA	PO ₄	T °C
S-501	Sh 4 purified liquor → X-501	liquid	34,133	183	81.8	1.2	43.3	0.2	0	0	28,497	0	121	4,962	229	0	40
S-502	NaOH 50 % → X-501	liquid	302	0	0	0	0	0	0	0	191	0	0	0	110	0	25
S-503	H2SO4 98 % → X-503 strip	liquid	135	0	0	0	0	0	0	0	2.7	0	132	0	0	0	25
S-504	X-503 → X-504 crystalliser	liquid	812	1.8	81.0	0	0	0	0	0	597	0	132	0	0	0	25
S-505	X-504/C-501/D-501 → CoSO4.7H2O product	solid	383	0	80.3	0	0	0	0	0	172	0	131	0	0	0	25
S-506	X-504 evaporator → condensate header	liquid	395	0	0	0	0	0	0	0	395	0	0	0	0	0	25
S-507	NaOH 50 % → X-511	liquid	414	0	0	0	0	0	0	0	263	0	0	0	151	0	25

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	NA	PO ₄	T °C
S-508	H2SO4 98 % → X-513 strip	liquid	297	0	0	0	0	0	0	0	5.9	0	291	0	0	0	25
S-509	X-513 → X-514 crystalliser	liquid	1,779	178	0	0	0	0	0	0	1,310	0	291	0	0	0	25
S-510	X-514/C-502/D-502 → NiSO4.6H2O product	solid	791	177	0	0	0	0	0	0	325	0	289	0	0	0	25
S-511	X-514 evaporator → condensate header	liquid	925	0	0	0	0	0	0	0	925	0	0	0	0	0	25
S-512	X-511 raffinate → T-501 Li3PO4 precipitation	liquid	34,588	3.6	0.8	1.2	43.3	0.2	0	0	28,952	0	121	4,962	490	0	40
S-513	E-501 → condensate header	liquid	20,889	0	0	0	0	0	0	0	20,889	0	0	0	0	0	60
S-514	Na3PO4 → T-501	dry solid	357	0	0	0	0	0	0	0	0	0	0	0	150	207	25
S-515	F-501 → Li3PO4 product	solid	263	0	0	0	41.2	0	0	0	34.3	0	0	0	0	188	25
S-516	E-501 brine → Sh 6 effluent (bleed)	liquid	13,792	3.6	0.8	1.2	2.2	0.2	0	0	8,028	0	121	4,962	640	18.8	60
S-517	condensate header → Sh 2 (S-203) + Sh 1/3 make-up	liquid	22,209	0	0	0	0	0	0	0	22,209	0	0	0	0	0	45
S-518	X-504 mother-liquor purge → Sh 6	liquid	33.6	1.8	0.7	0	0	0	0	0	30.0	0	1.1	0	0	0	25
S-519	X-514 mother-liquor purge → Sh 6	liquid	63.8	1.4	0	0	0	0	0	0	60.0	0	2.4	0	0	0	25
S-520	F-501 filtrate → E-501 evaporator	liquid	34,682	3.6	0.8	1.2	2.2	0.2	0	0	28,917	0	121	4,962	640	18.8	40

NOTES – CIRCLED NUMBERS ON THE DRAWING

- (25) **Cobalt first, Cyanex 272.** The industry Co/Ni separation. Tait ^[15] measures pH₅₀ ≈ 5.3 for Co and ≈ 7.0 for Ni, so at pH 5.5–6.0 cobalt loads and nickel mostly stays. The window is real but narrower than the 4.5 / 6.5 pair often quoted in reviews. Saponified organic (NaOH pre-neutralised) so pH does not crash on loading; the sodium that adds joins the barren brine. Three extraction stages for 99 %, two scrub stages to hold Ni in the product below spec. Gluconate in the aqueous is a weak Co ligand (log K ~1.9) and does not compete with the extractant.
- (26) **Nickel by Versatic 10, not a second Cyanex circuit.** Cyanex 272 takes Ni only near pH 7 where hydroxide risk starts; Versatic 10 at pH 6.5–7 is the standard Ni extractant over Li/Na/Ca. Strip with H₂SO₄ to a NiSO₄ liquor for crystallisation — the sulfate matrix for the product is made here, not inherited from the leach (this is a gluconate plant).
- (27) **Products are the hydrated sulfates.** CoSO₄·7H₂O and NiSO₄·6H₂O at **86 % per-pass** crystalliser yield ^[17], with the mother liquor **recycled to the crystalliser feed** and only a 5 % bleed leaving as S-518/519 to Sheet 6. Because the bleed rather than the per-pass yield sets the loss, overall crystalliser recovery is 99.2 %. Overall recovery to saleable product: **Co 80 %, Ni 78 %, Li 94 %** — the "downstream losses applied" answer to the 86 / 84 / 100 % leach figures. Lithium is recovered as **Li₃PO₄** rather than Li₂CO₃; 1,831 t/yr at 94.1 % Li recovery. Products are 3,065 / 6,324 / 1,831 t/yr of CoSO₄·7H₂O, NiSO₄·6H₂O and Li₃PO₄.

- 28 **The evaporator recovers water, not lithium.** The barren liquor (S-512, ~23 t/h) is already ~240 g/L sodium gluconate, so it can never be concentrated 13× to a carbonate-grade 25 g/L Li — the salt would pass its ~590 g/L solubility long before. Lithium therefore comes out first as Li_3PO_4 (note 29) and E-501 then concentrates the Li-free filtrate (S-520) only to ~500 g/L sodium gluconate: ~9 t/h of water evaporated, returned as clean condensate (S-517 with the crystalliser condensates, ~10 t/h) to the fermenter medium and make-up on Sheets 1–3. MVR keeps this to roughly 25–35 kWh per tonne evaporated. Alternatives recorded and not taken: recycling the raffinate to R-301 (`recycle_loop_check.py`: it returns as sodium gluconate at pH 6.5 with no free acid) and bipolar-membrane electrodialysis to split the salt back into acid + base (the true closed loop; a Sheet 6 option).
- 29 **Li_3PO_4 at 60 °C with trisodium phosphate.** Li_3PO_4 solubility is ~0.39 g/L (0.07 g/L as Li), two orders below Li_2CO_3 , so the barren at ~1.9 g/L Li precipitates directly: $3 \text{Li}^+ + \text{PO}_4^{3-} \rightarrow \text{Li}_3\text{PO}_4$. Na_3PO_4 at 10 % excess (~355 kg/h), 95 % yield assumed (residual ~0.1 g/L Li), 1 h residence, water wash on F-501; the excess phosphate leaves in the brine. Product 228 kg/h (1,821 t/yr) is a lower-value intermediate than battery-grade Li_2CO_3 — it is converted to Li_2CO_3 or LiOH by the buyer ($\text{H}_3\text{PO}_4/\text{Ca}$ or electrochemical routes) — and is priced accordingly. Adds ~150 kg/h Na to the brine.
- 30 **The brine is the bleed.** S-516 (~13.7 t/h at 60 °C, ~500 g/L sodium gluconate) carries essentially all the sodium (~630 kg/h from three NaOH additions plus the phosphate — set by pH targets in the gluconate buffer), ~4.9 t/h of gluconate, excess phosphate, organics, and residual Li/Co/Ni. Nothing salty goes back to the fermenter. It is the main COD (~4.4 t/h as O_2 , gluconate ThOD ≈ 0.9 g/g) and salt duty on Sheet 6 — and the reason NaOH dosing is the number to minimise across Sheets 4–5. Technical-grade sodium gluconate (concrete admixture) is a possible outlet if phosphate and metals are controlled; food/pharma grades are not realistic from this stream.

Block 7 Area 600 · Effluent and by-product (Sheet 6)

EQUIPMENT

TAG	SERVICE	TYPE	MATERIAL / NOTES
T-601	Effluent equalisation	Agitated tank, 8 h	Receives the E-501 brine (S-516), both crystalliser purges (S-518/519) and the SC-301 scrubber blowdown (S-304). FRP or rubber-lined; ~50 °C (note 31)
D-601	Crude-salt dryer	Steam-tube rotary or spray dryer	Dries the equalised brine to a 5 % moisture crude sodium-gluconate salt (S-603). Exhaust S-602 is humid air — a condenser here would recover ~7.8 t/h and make the plant water-neutral (note 32)
X-601	Leach residue bunker	Covered concrete bay, 3 days	S-308 graphite/binder/oxide residue with the Fe/Al that precipitated in the leach; to graphite recovery or licensed disposal (note 33)
X-602	Fe/Al hydroxide cake bunker	Covered bay, 3 days	S-408 goethite/gibbsite cake at 60 % moisture, carries the 2 % Co/Ni occlusion loss
X-603	Spent-biomass skip	Skip, 7 days	S-210 G. oxydans cells from F-201; compostable
Cu cement	By-product	Drummed	S-403 cement copper (~19 kg/h Cu with excess Fe) to a smelter — leaves from Sheet 4, listed here for completeness

SIZING — GENERATED BY MASS_BALANCE.PY

TAG	SERVICE	SIZE	BASIS
T-601	Effluent equalisation tank	112 m3	8 h of combined liquid effluent
D-601	Crude-salt dryer (steam-tube / spray)	7.9 t/h H2O	water evaporated from the brine
X-601	Leach residue bunker	80.7 t	3 days of wet cake
X-602	Fe/Al cake bunker	37.6 t	3 days of wet cake
X-603	Spent-biomass skip	6.7 t	7 days of wet cake

STREAMS (KG/H) — GENERATED BY MASS_BALANCE.PY

NO.	FROM → TO	PHASE	TOTAL	NI	CO	MN	LI	CU	AL	OTHER	H ₂ O	FE	SO ₄	GLUCONATE	NA	PO ₄	T °C
S-601	T-601 equalisation → D-601 dryer	liquid	13,981	6.9	1.5	1.2	2.2	0.2	0	0	8,168	0	124	4,962	640	18.8	50
S-602	D-601 exhaust → atmosphere (humid air)	gas	7,862	0	0	0	0	0	0	0	7,862	0	0	0	0	0	90
S-603	D-601 → crude Na-gluconate salt, OSBL	solid	6,119	6.9	1.5	1.2	2.2	0.2	0	41.6	306	0	124	4,962	640	18.8	40

NOTES

- 31
- One liquid effluent, and it is dried, not discharged.** The plant's only liquid bleed is the sodium/phosphate/gluconate stream: ~13.9 t/h combined (brine 13.7, purges 0.1, scrubber blowdown 0.1). Treating it biologically would be a 4.4 t/h COD (as O₂) duty — a wastewater plant bigger than the leach — and would throw away 4.9 t/h of gluconate that cost 4.8 t/h of glucose to make. Drying it instead gives ~6.1 t/h (49 kt/yr) of crude sodium gluconate (~90 % Na-gluconate, plus Na₂SO₄, excess phosphate, carbonate from the scrubber, and ppm Co/Ni/Li). Technical-grade outlets (concrete set-retarder, industrial chelating cleaners) exist at roughly \$0.6–0.9/kg; food or pharma grade is not reachable from this stream without a separate purification plant. **The salt is carried at zero credit:** metals recovery is the purpose of the plant, so it is an off-take obligation rather than a revenue line, and the glucose behind it is an unoffset cost. Any off-take actually signed is upside on the metals case. *The alternative — bipolar-membrane electrodialysis returning gluconic acid to R-301 and NaOH to Sheets 4–5 — is the true closed loop and is the option to study next.*

- 32 **Water closes with a small fresh make-up.** Condensate (crystallisers + E-501, ~10.4 t/h) supplies the S-104 make-up and most of the fermenter medium; the remainder of the medium (~5.8 t/h) and the F-301 wash (~1.6 t/h) are fresh water, less a 0.5 t/h condensate surplus — net ~6.9 t/h fresh. The D-601 exhaust carries ~7.8 t/h of water: condensing it would make the plant water-neutral, at the cost of a condenser and a slightly larger dryer. Water leaves in the product hydrates (~1.0 t/h), the three cakes (~0.7 t/h), the salt (0.3 t/h) and the dryer exhaust. The fermenter off-gas humidity is not counted.
- 33 **Solids leave as three cakes.** Leach residue S-308 (~1.1 t/h wet; graphite, binder, unleached oxide and the goethite/gibbsite that precipitates in R-301 at pH ~4.8), Fe/Al cake S-408 (~0.5 t/h wet), spent biomass S-210 (40 kg/h). None is hazardous by composition, but the residue carries fluoride from the electrolyte (not modelled) and should be leach-tested before a disposal route is fixed. Copper cement S-403 is a by-product, not a waste.

References IEEE style

Sources verified by DOI resolution where possible. Entries marked in bold could not be confirmed word-for-word because the publisher blocked full-text access; the paper and its identity are real, but the author list or exact figures should be checked against the PDF before print. Ordinary engineering practice (tank residence times, pump sparing, silo hold-up) is deliberately left uncited. Entries [1]–[4] are cited in the write-up; the numbering matches K–references/REFERENCES.md.

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- [27] Z. Wang *et al.*, "Recycling technologies for cathode materials from spent lithium iron phosphate batteries: an overview," *Materials*, vol. 19, no. 4, Art. no. 674, Feb. 2026, doi: 10.3390/ma19040674. **[bibliographic details confirmed via Crossref. Note this is a review: the Li₃PO₄ figures it carries are quoted from its own sources, not an original assessment, so they inherit that provenance.]**
- [28] Fastmarkets, "Cobalt sulfate 20.5% Co basis, exw China, yuan/tonne," assessment MB-CO-0017. [Online]. Available: <https://www.fastmarkets.com/commodity-prices/> (subscription required; series confirmed, value not public).

ASSUMPTIONS CARRIED WITHOUT A CITED VALUE

- **Fe(III)–gluconate stability constant.** Not fixed. Sawyer ^[16] is the reference to consult; values from 8 to 10 change whether iron precipitates cleanly or stays complexed.
- **Li₃PO₄ solubility 0.39 g/L**, and its behaviour at 60 °C. Sets whether lithium precipitates from the 1.9 g/L barren without evaporation.
- **Co/Ni occlusion in the hydroxide cake, 2 %.** Literature suggests 2–5 %; no specific figure verified.
- **MVR specific energy 25–35 kWh per tonne evaporated.** Order of magnitude is right; the range is vendor-typical.
- **Solvent-extraction stage counts** (3 extraction / 2 scrub / 2 strip) and **0.2 µm air filtration**. Ordinary design practice, uncited by intent.
- **Gluconic acid pK_a 3.86.** Sources also report 3.70 ^[20]; the spread is the equilibrium with glucono-δ-lactone. Under 2 % effect on base demand.
- **Every price.** Battery-grade sulfate assessments are subscription-only ^[28] and no agency assesses Li₃PO₄ ^[27], so revenue is a floor on contained metal at USGS benchmarks ^[24, 25, 26]. All cost inputs are named constants in I–cost–estimate/. Full list: J–unsourced–assumptions/.